

Agentic Coding IDE Extensions Introduction



SoftUni Team
Technical Trainers



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<https://softuni.org>

Have Questions?

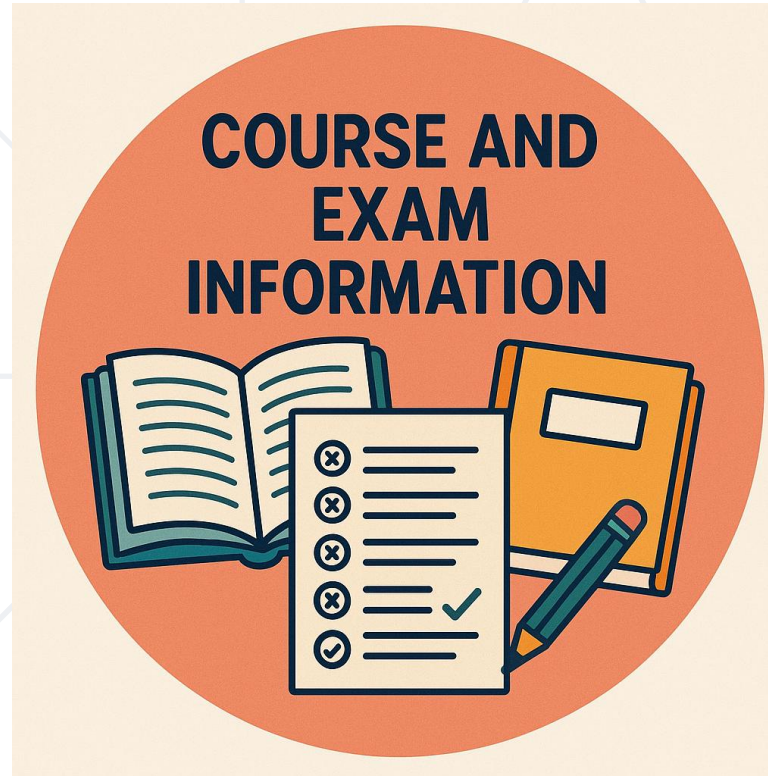
sli.do

#AI-Dev

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Course and Exam Information

Course Information

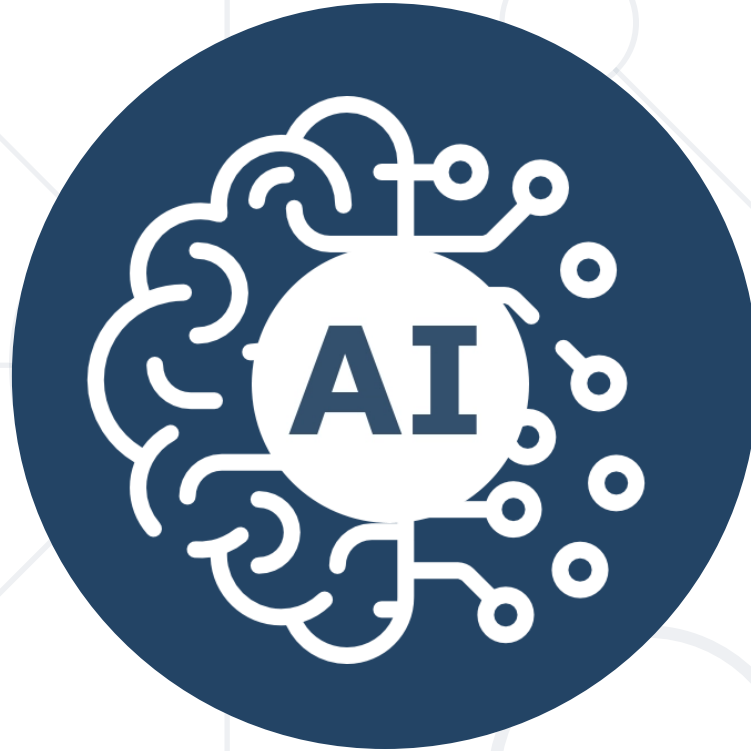
1. Not focused on any specific language
2. Demos will use multiple languages
3. Practices work across all tech stacks



Exam Project

- Create a project on your own topic
- Start now
- Submit until the end of the course

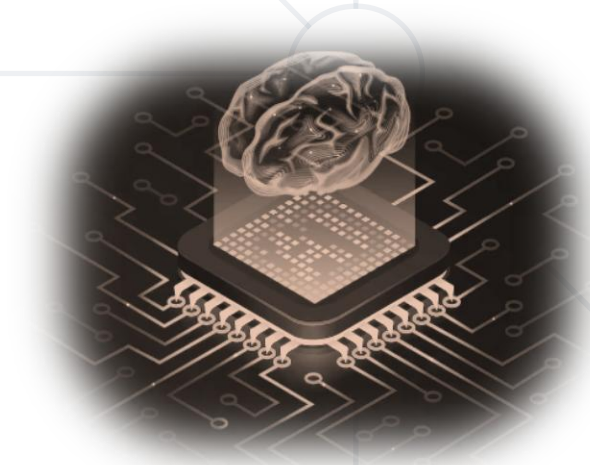




Generative AI and LLMs

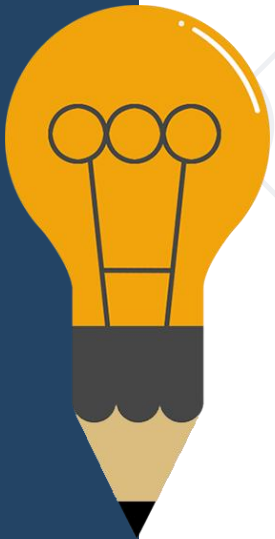
LLMs, Coding Models, Agents, Tokens

- **Artificial intelligence** (AI) == intelligent computer system
 - Performs tasks that typically require **human intelligence**
- **Machine learning**
 - AI uses **algorithms** and mathematical **models** that are trained on huge **data sets**, through many, many **examples**
- Examples of **machine models** (AI models):
 - **GPT-5** generates text (by text prompt)
 - **FLUX 1.1** generates images (by text prompt)
 - **SpeechT5** generates human speech (by text)



Generative Artificial Intelligence

- **Generative Artificial Intelligence** (GenAI)
 - **Large machine models**, trained on human text, images, documents, audio, video
 - **Generate** text, code, images, videos by **prompt**, answer questions, solve problems, create software
 - **Modalities** (what models take as input and output): text, images, screenshots, audio, video, code
 - Require **huge computing resources** for training and a significant resources for inference (generating answers)



- **Large Language Models (LLMs)**
 - **Text content generators**, trained on huge data sets (books, articles, websites, programming code, synthetic data, etc.)
 - Example: **Llama 4 Maverick** is trained on **22T tokens**
 - Learn dozens of **languages**, grammar, text writing and **coding**
 - Fine-tuned to **answer questions** and **follow instructions**
 - Have **parameters** (capacity to remember information and relationships), e. g. **32B**, **405B** or **1.4T** parameters
 - Examples: ChatGPT, Claude, Gemini, DeepSeek, Llama, Mistral

- **Tools** are external commands / services used by LLMs to perform tasks beyond text generation
 - For example, using tools, LLM can run **terminal commands**
 - ChatGPT uses tools to search Internet and draw charts
- **Reasoning models** (thinking models) simulate human-like logical thinking, planning and problem-solving
- **AI dev agents** use tools + reasoning to perform multi-step tasks
 - First, the agent **creates a plan**, using a reasoning model
 - Then, it **executes the plan** step-by-step: writes code, uses tools, runs the code, checks for errors, fixes bugs, etc.



Intro to AI-Assisted Development

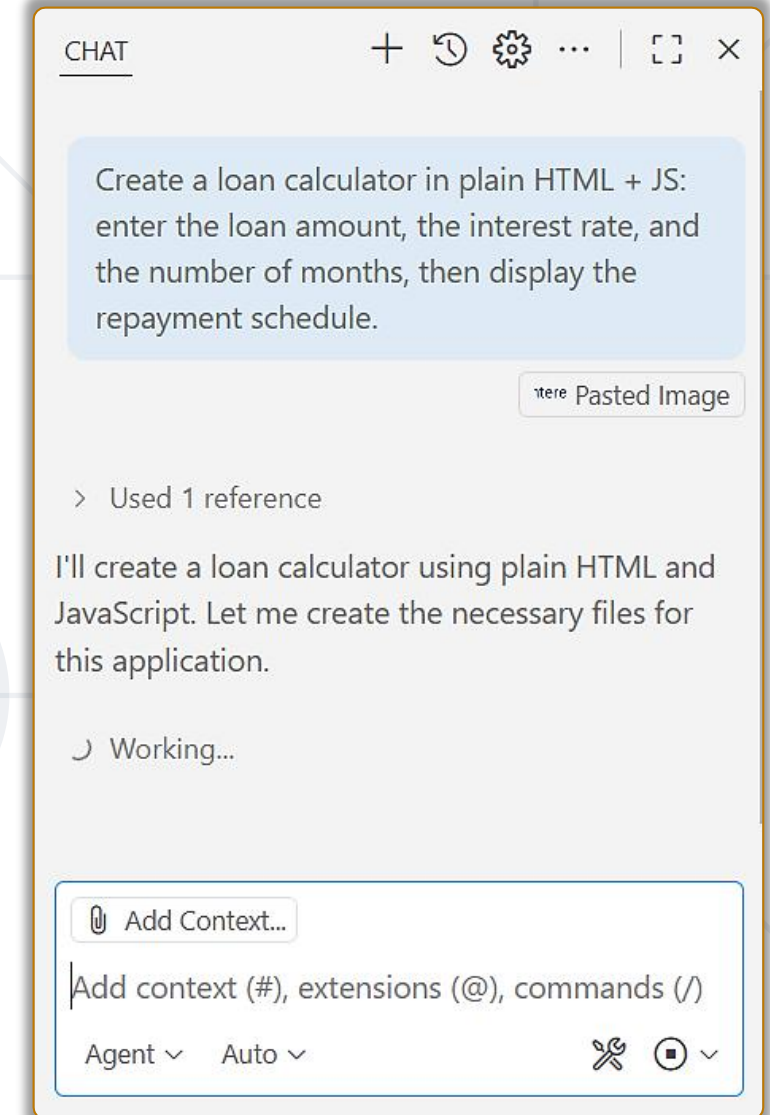
App Development with AI Prompts

What is AI-Assisted Development?

- Create apps using **AI prompts**
 - You write a sequence of **AI prompts** (like in ChatGPT)
 - **AI writes the code** for you, runs, tests, fixes bugs, etc.
- AI-assisted dev environments
 - For **hobbyists**: **Bolt, Lovable, Replit, v0, Firebase Studio**
 - Quite simplified, cloud-based, running in the browser
 - For **professionals**: **GitHub Copilot, Cursor, Claude Code**
 - Installed locally, more complex, quite more powerful



- **GitHub Copilot (+ VS Code)**
 - Professional **AI dev environment** with powerful **AI dev agent**
 - Natively **integrated into VS Code** (the world's most widely used IDE)
 - **Huge popularity**: 68% of developers use Copilot, 20M users ([Sept 2025](#))
 - Chat modes: **Agent** / **Ask** / **Plan**
 - <https://github.com/features/copilot>



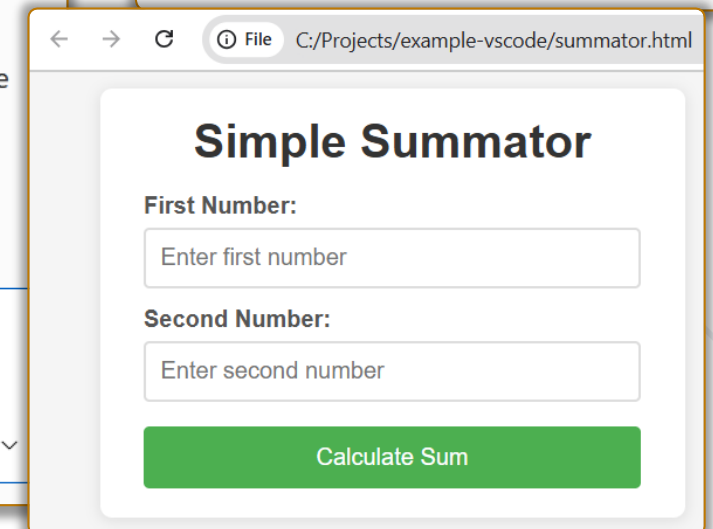
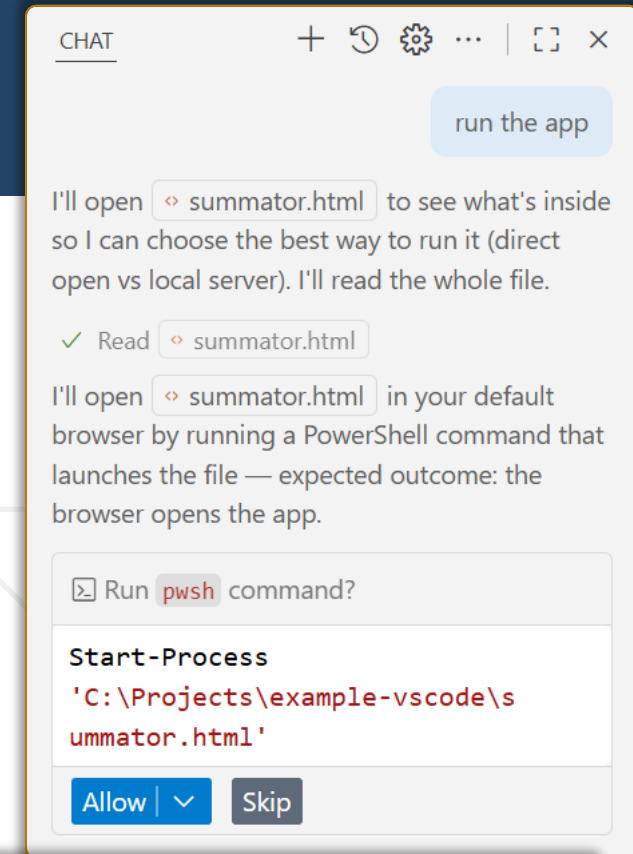
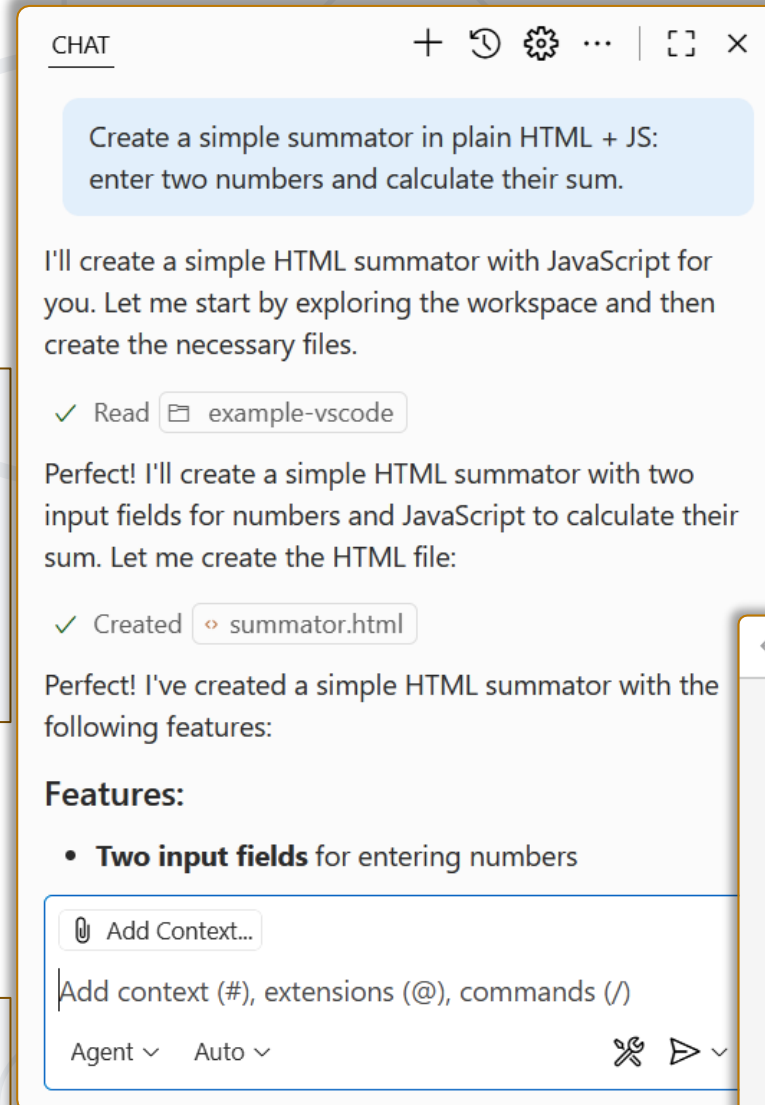
GitHub Copilot – Live Demo

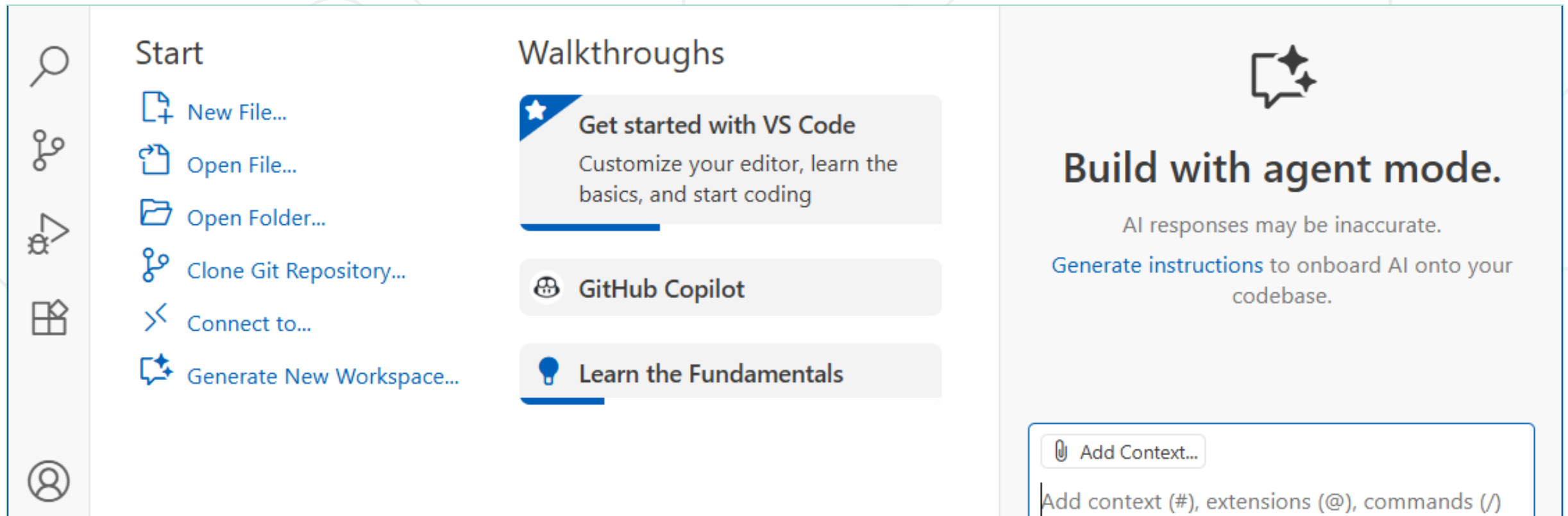
- Open VS Code
- Type this **prompt** in GitHub Copilot:

Create a **simple summator** in plain **HTML + JS**: enter two numbers and calculate their sum.

- Wait for the app to be generated → **start** it

run the app



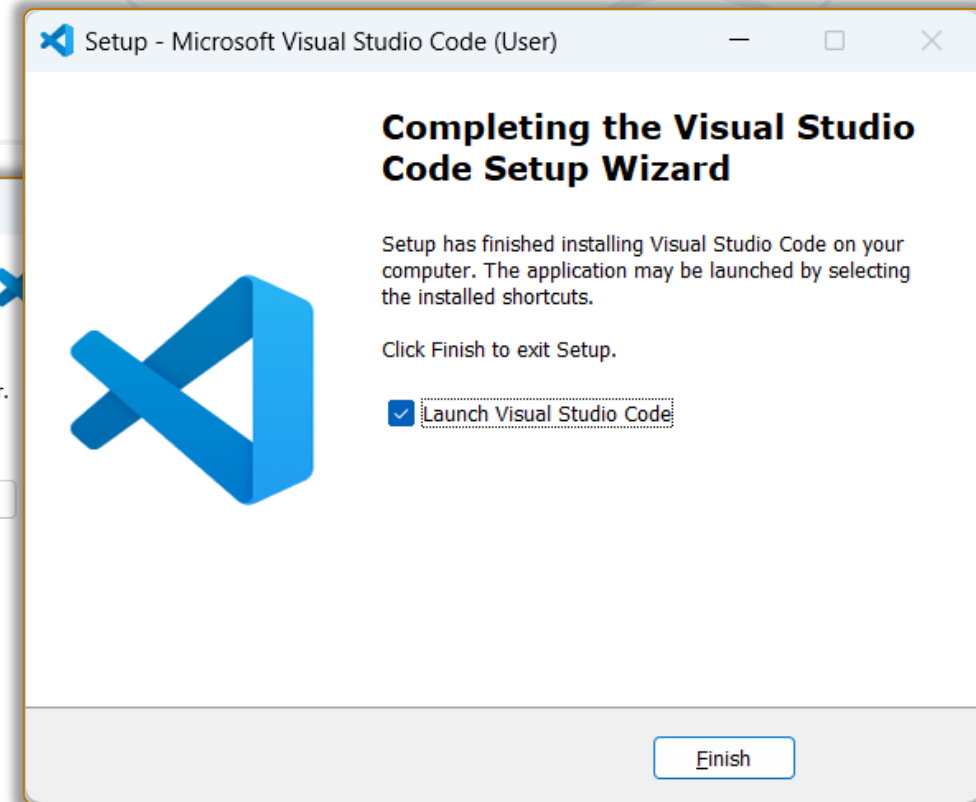
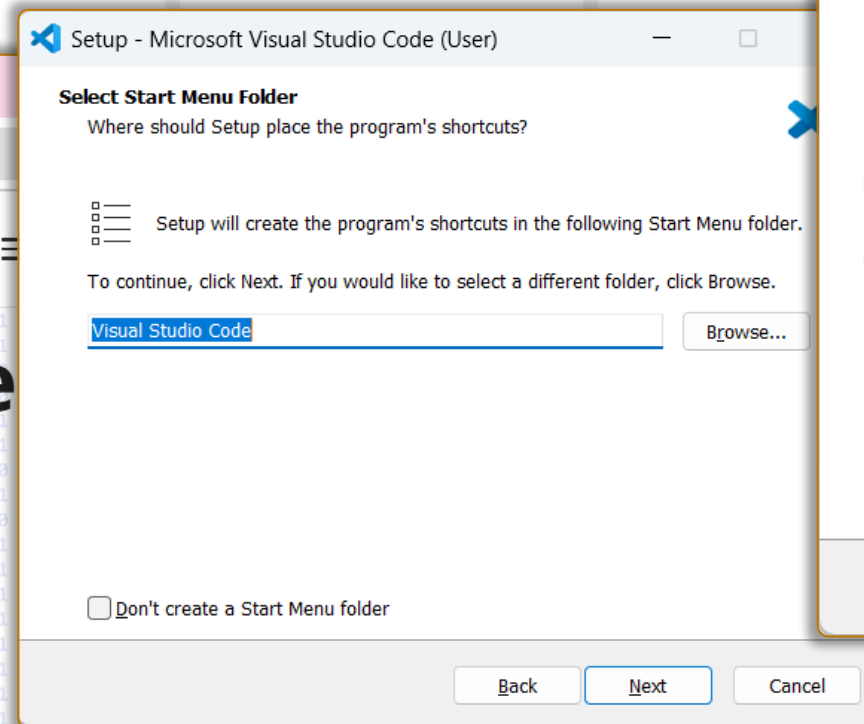
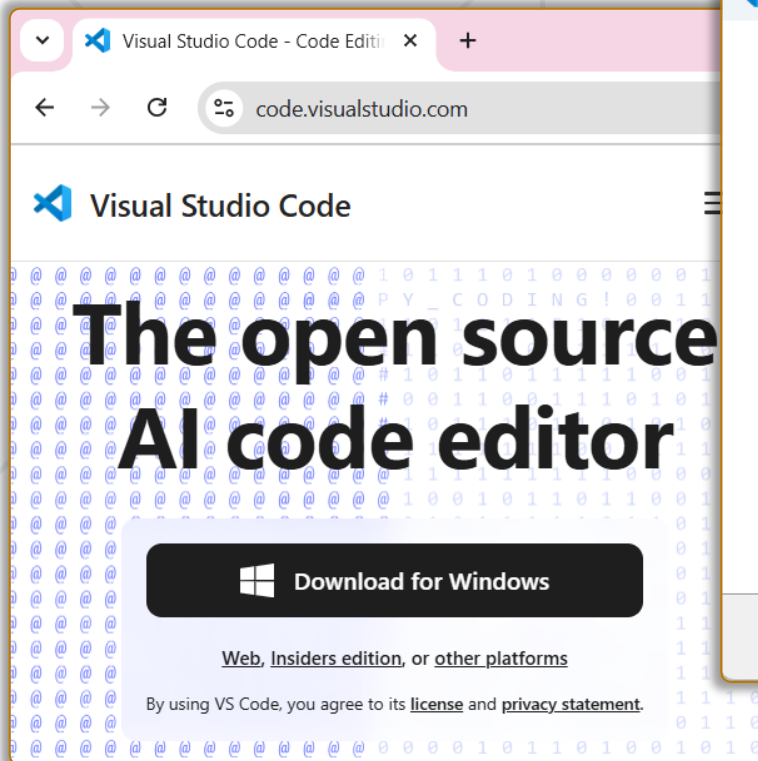


Installation, Alternatives, Pricing

VS Code + Extensions

Installing Visual Studio Code

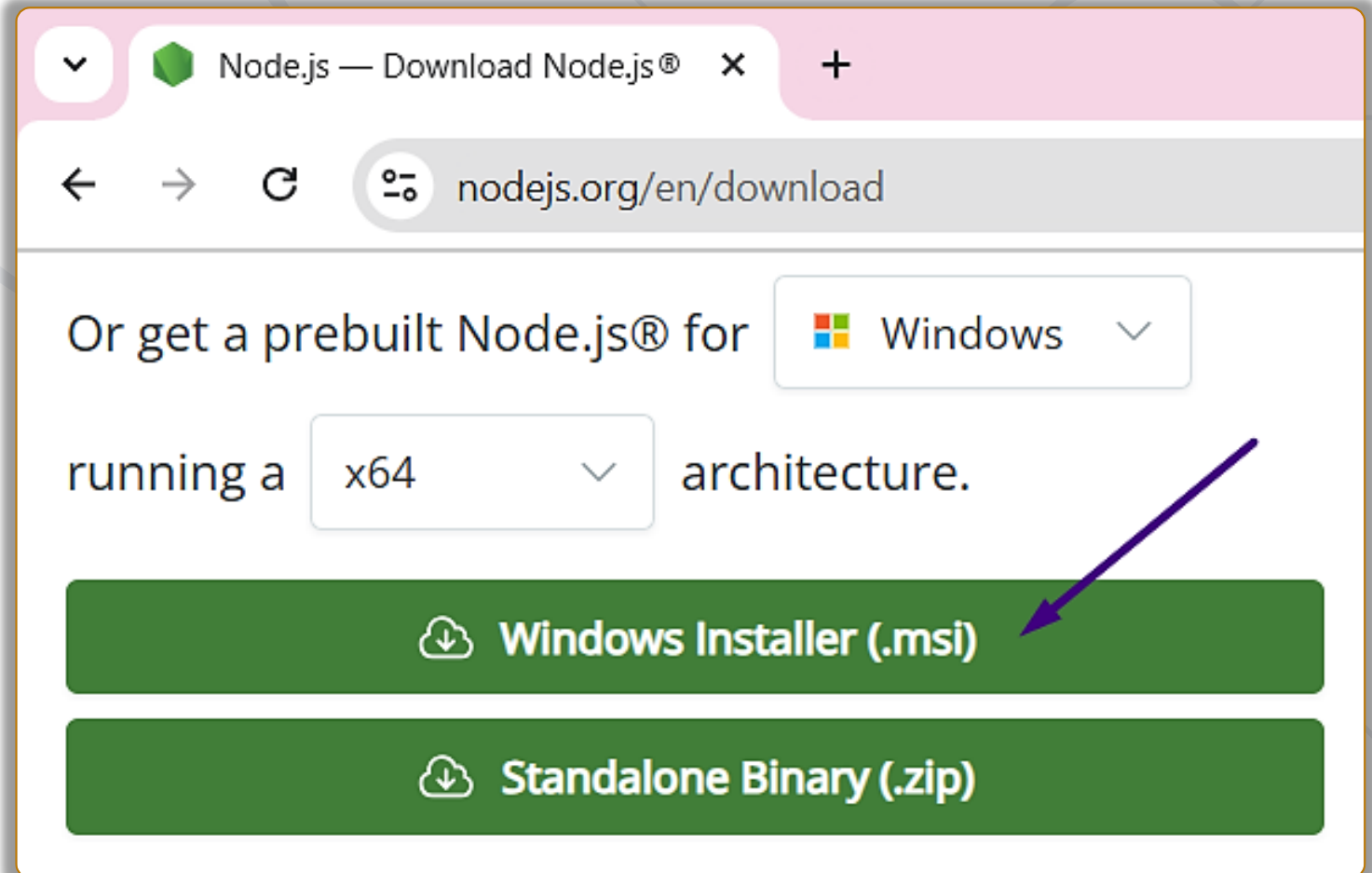
- **Download and install VS Code** (Visual Studio Code) from:
 - <https://code.visualstudio.com>



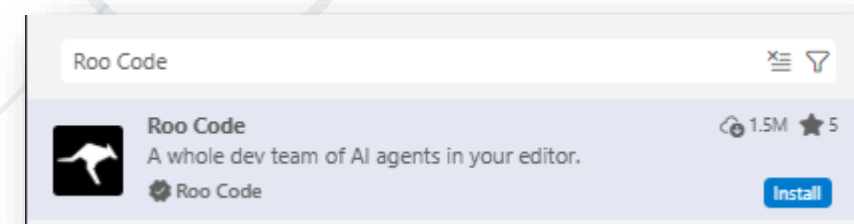
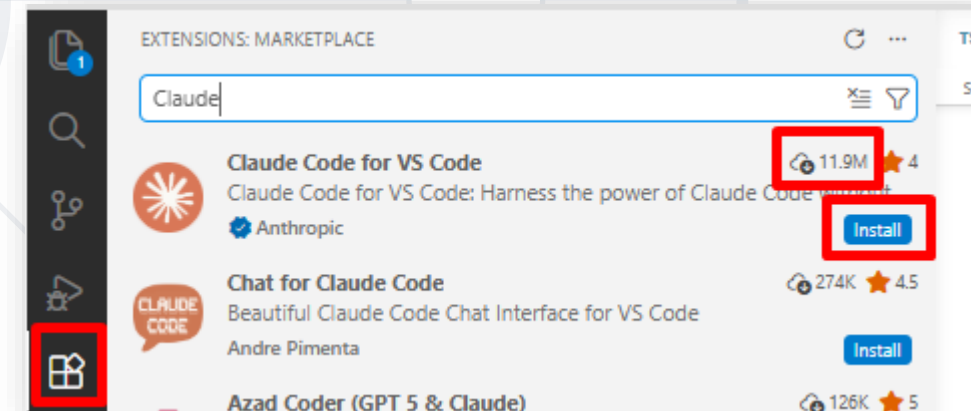
Install Node.js



- **Node.js** is a local JavaScript runtime for developers
- <https://nodejs.org>



- All **extensions** provide 90%+ **similar** functionality
- You can pick **any** option or **use multiple** alternatives at the same time
 - GitHub Copilot
 - Claude
 - Codex
 - Roo Code
 - Augment Code
 - Other alternatives



Licensing & Pricing

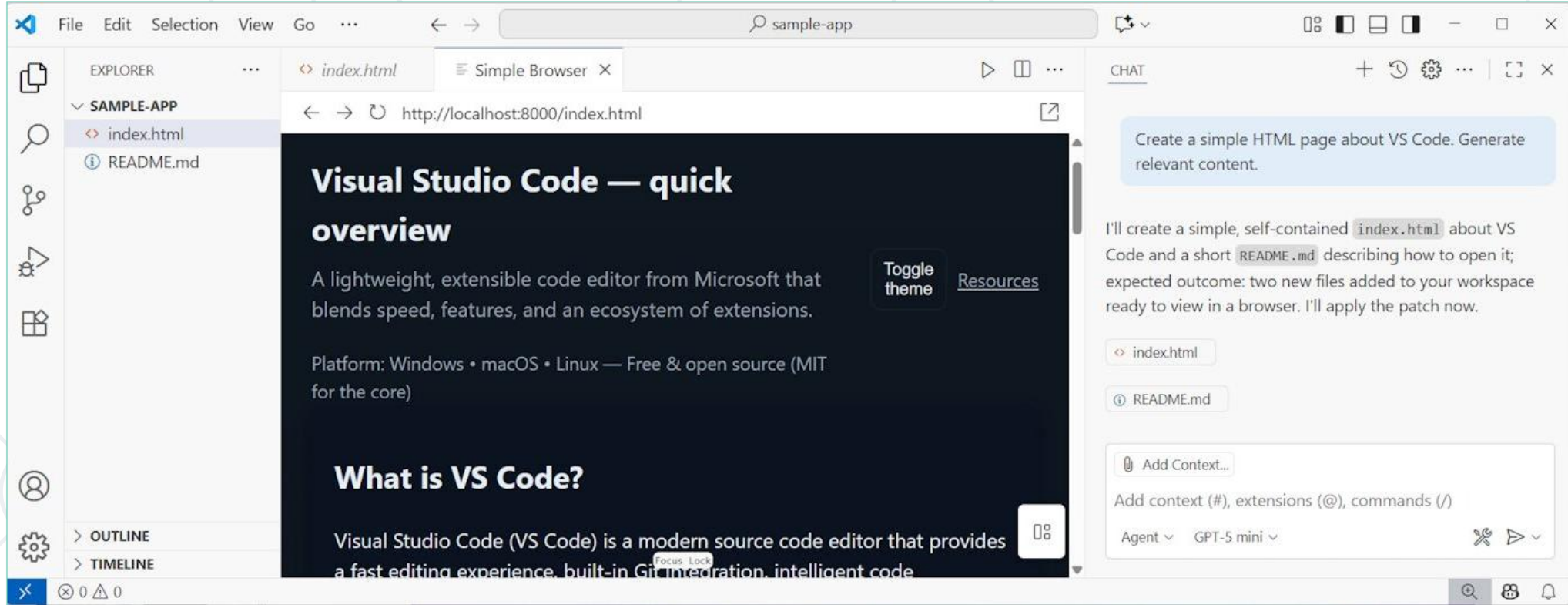
- **Paid** accounts with Free Tier
- **Free** extension – **paid models** (via OpenRouter or provider)
- Free extension with limited free models
- Advice: Start with free, connect via **OpenRouter**
- Constant changes
 - [LINK 1](#)
 - [LINK 2](#)

Free	Pro Most popular	Pro+
A fast way to get started with GitHub Copilot.	Accelerate workflows with GitHub Copilot.	Scale with agents and more models.
\$0 ^{USD}	\$10 ^{USD} per user / month	\$39 ^{USD} per user / month
Get started	Temporarily unavailable - read more	Temporarily unavailable - read more

The Unified Interface For LLMs

Better **prices**, better **uptime**, no subscriptions.

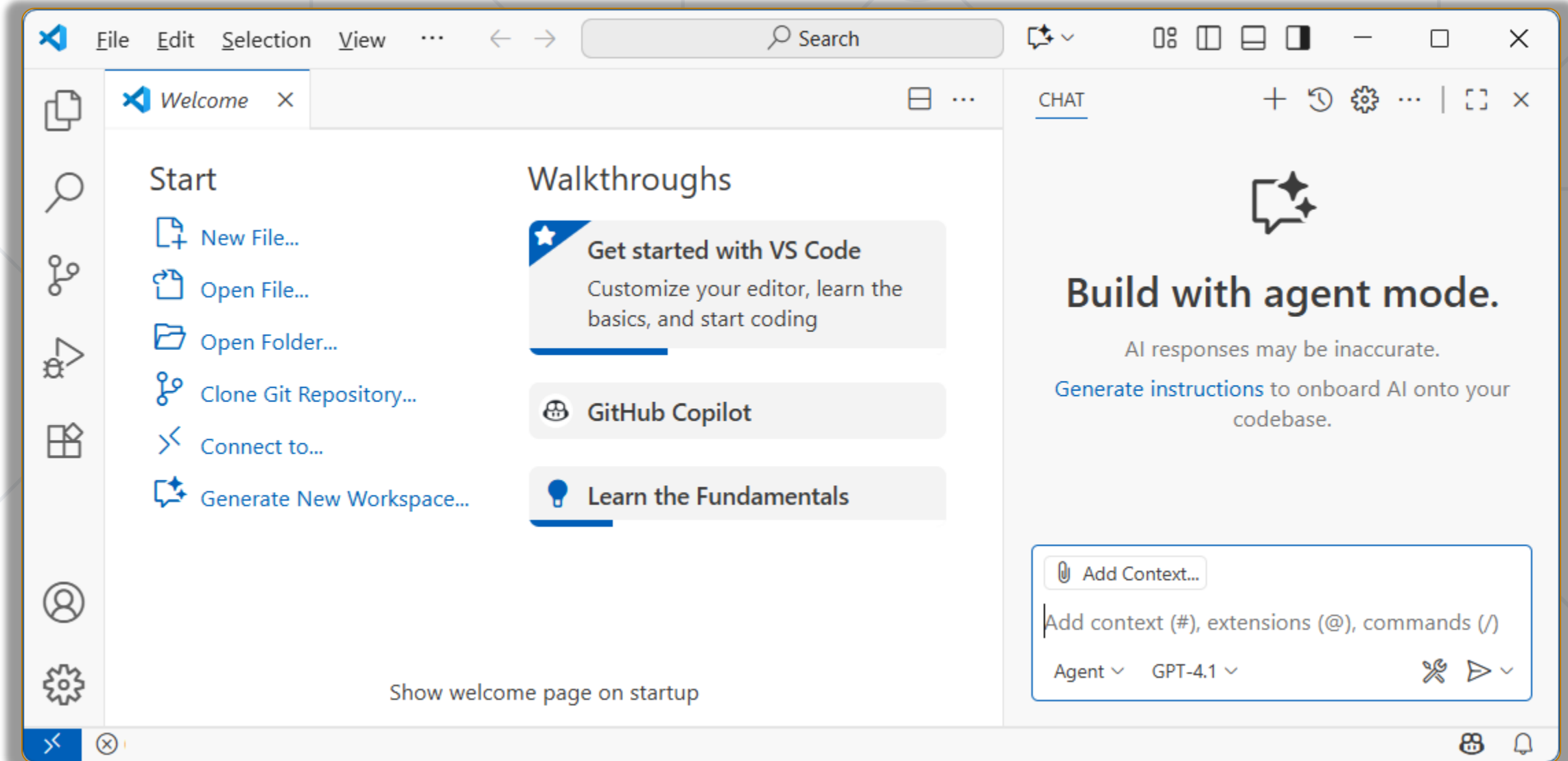
[Get API Key](#)[Explore Models](#) 



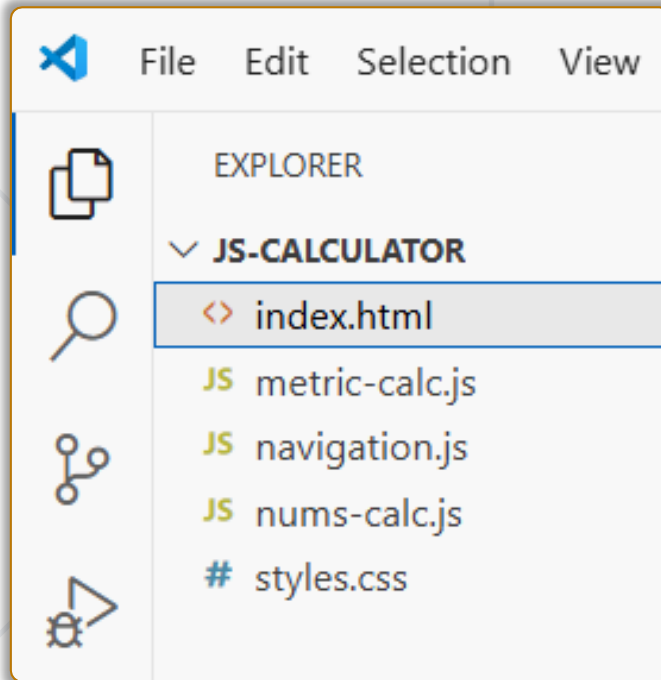
Your First App with AI-Assisted Coding

- **VS Code** is modern development IDE  Visual Studio Code
 - In VS Code, developers **write** code, **execute** the code, **debug** and bug fix the code, **test** the code, bundle and **deploy** apps
 - **Extensions** to support multiple languages / technologies
 - HTML, CSS, JavaScript, Python, C#, Java, PHP, C++, SQL, ...
 - Docker, Azure, AWS, Markdown, Jupiter, Supabase, ...
 - Integrated **GitHub Copilot** for AI-assisted coding
 - <https://code.visualstudio.com>

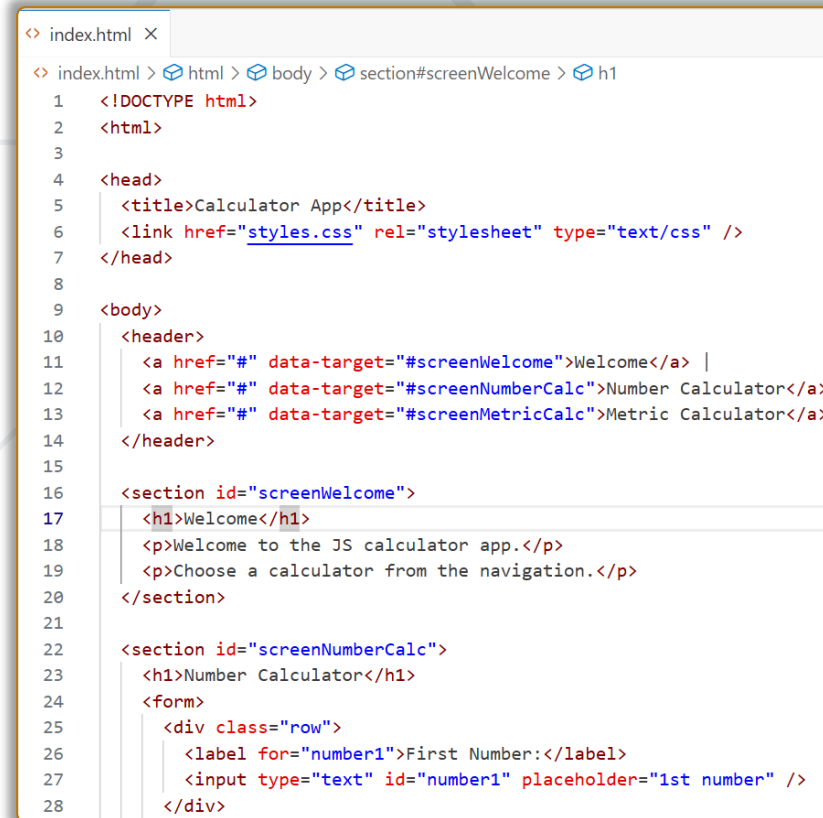
VS Code with GitHub Copilot



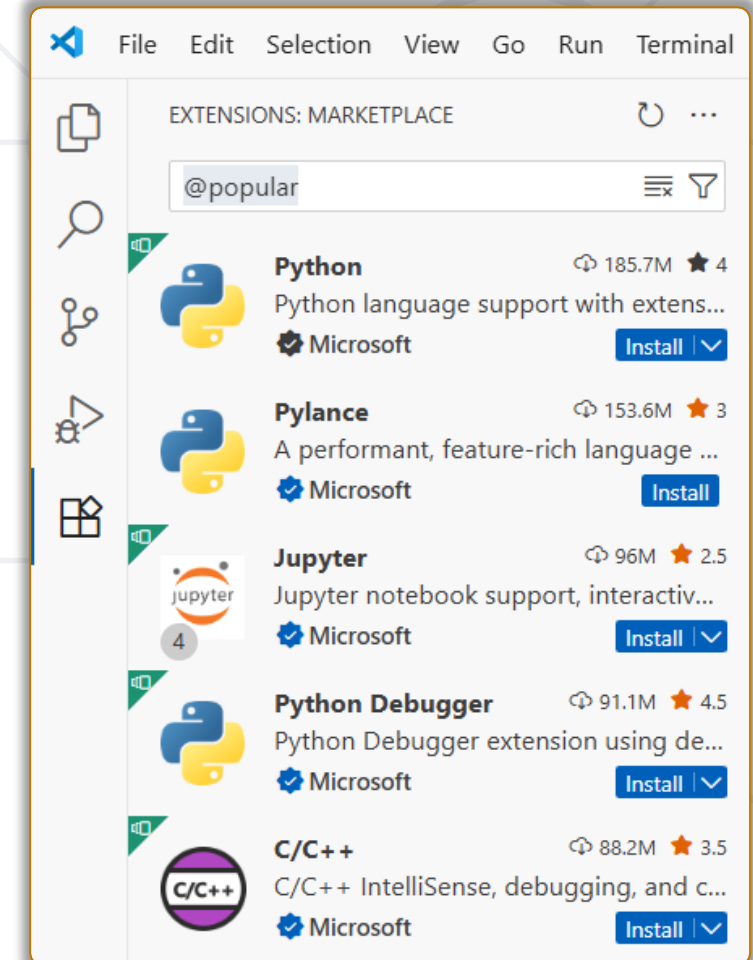
Code explorer



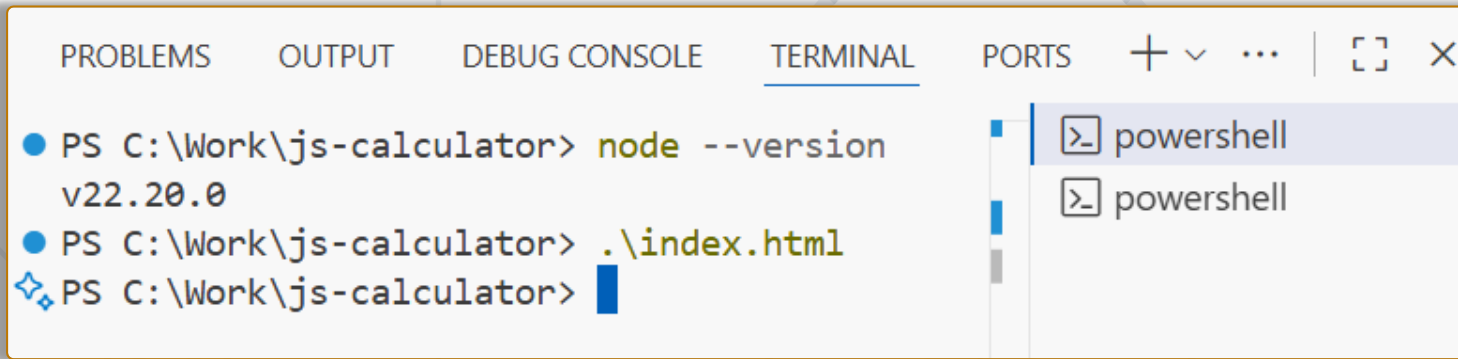
Code editor



Extensions



- **Terminal** for running command-line tools

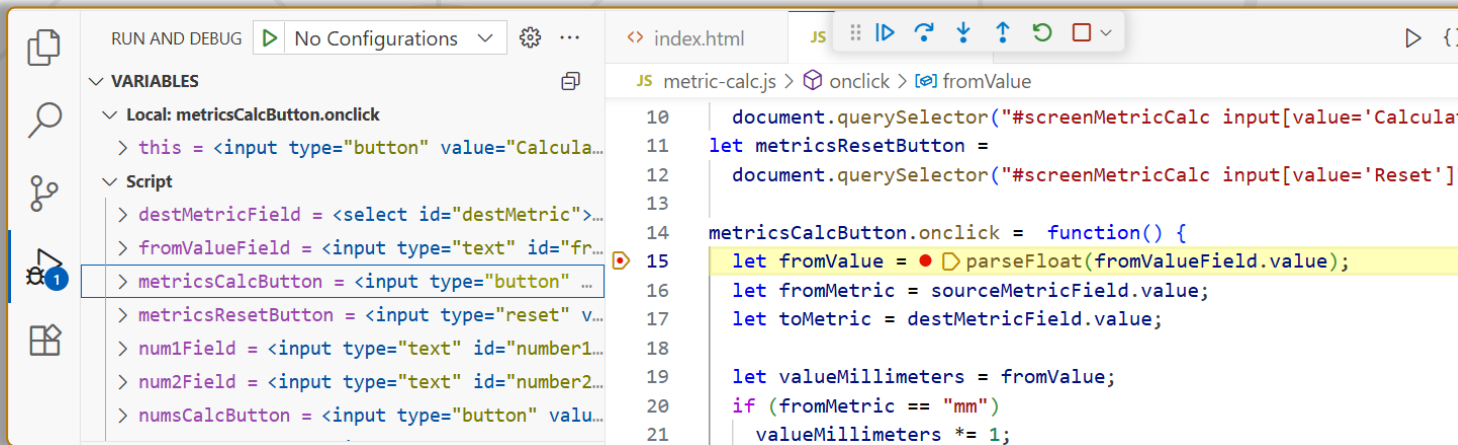


The screenshot shows the VS Code interface with the 'TERMINAL' tab selected. The terminal displays the following commands and output:

```
PS C:\Work\js-calculator> node --version
v22.20.0
PS C:\Work\js-calculator> .\index.html
PS C:\Work\js-calculator>
```

On the right side of the terminal window, there is a list of terminal instances, both labeled 'powershell'.

- **Debugger** for tracing code execution and finding bugs



The screenshot shows the VS Code interface with the 'RUN AND DEBUG' tab selected. The left sidebar displays the 'VARIABLES' panel, showing the state of variables during execution:

- Local: metricsCalcButton.onclick**
 - `this` = `<input type="button" value="Calcula..."`
- Script**
 - `destMetricField` = `<select id="destMetric">...`
 - `fromValueField` = `<input type="text" id="fr..."`
 - `metricsCalcButton` = `<input type="button" ...`
 - `metricsResetButton` = `<input type="reset" v...`
 - `num1Field` = `<input type="text" id="number1...`
 - `num2Field` = `<input type="text" id="number2...`
 - `numsCalcButton` = `<input type="button" valu...`

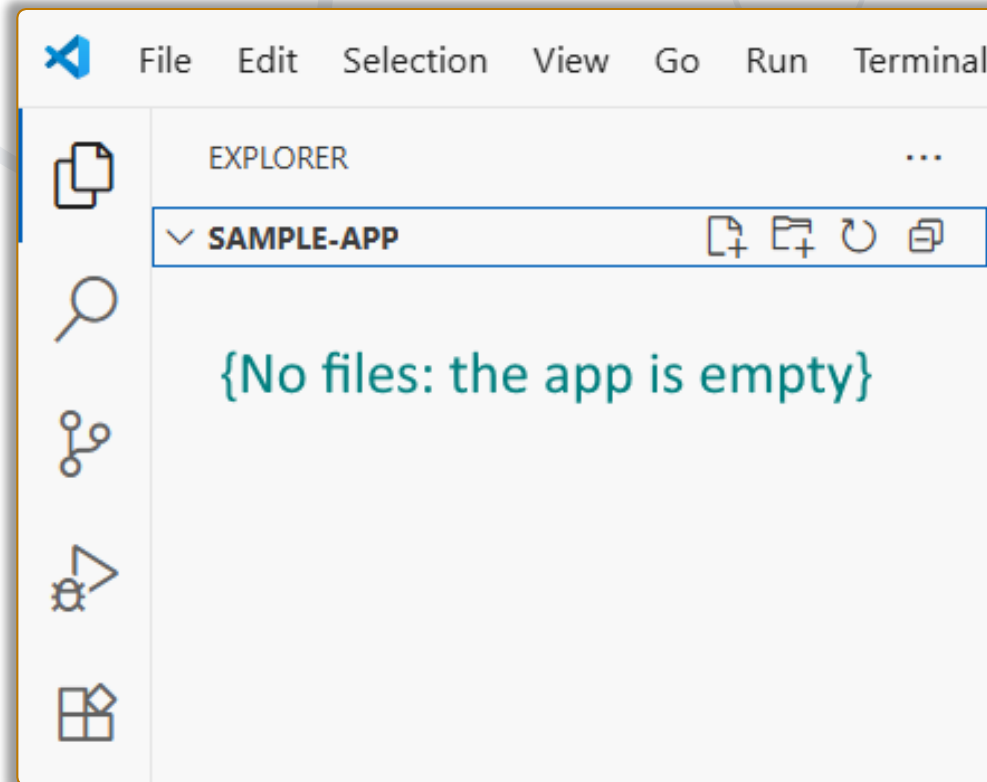
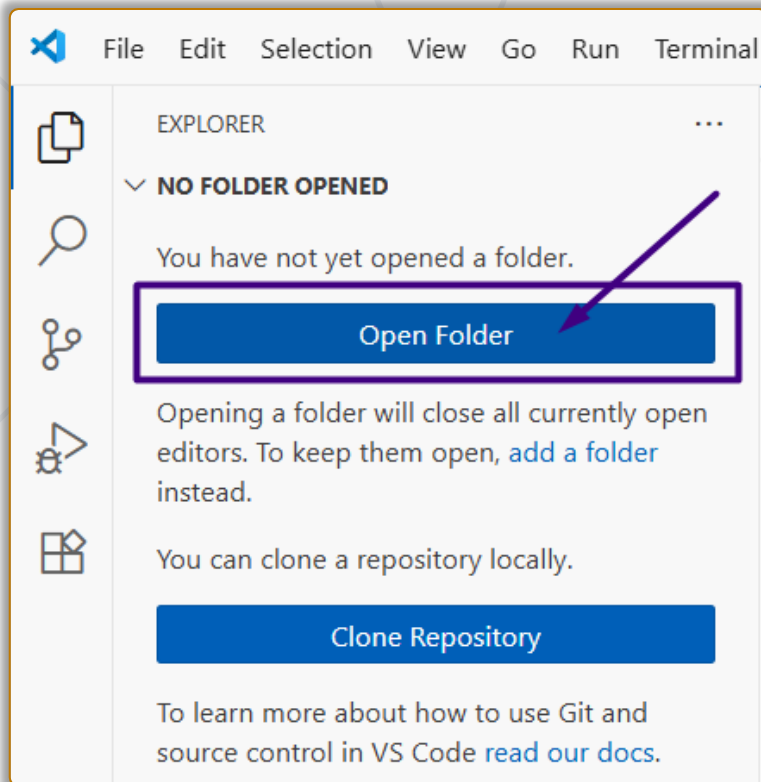
The main editor shows the JavaScript code for `metric-calc.js`. The execution is paused at line 15, which is highlighted in yellow:

```
15 let fromValue = parseFloat(fromValueField.value);
```

The code continues with calculations and updates to the UI elements.

Creating a New Project

- In VS Code **projects** are structured in **folders**
- Create a **new folder** and **open the folder** to start coding

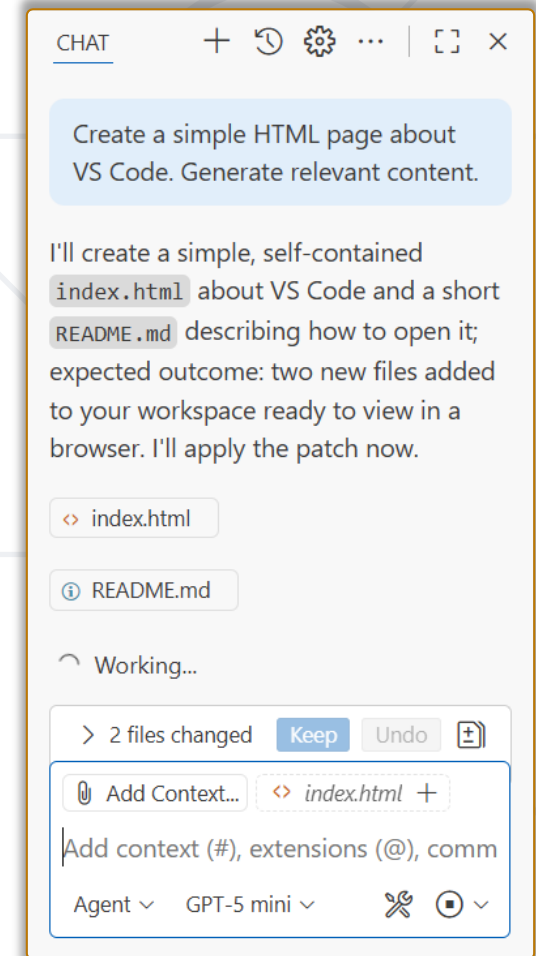
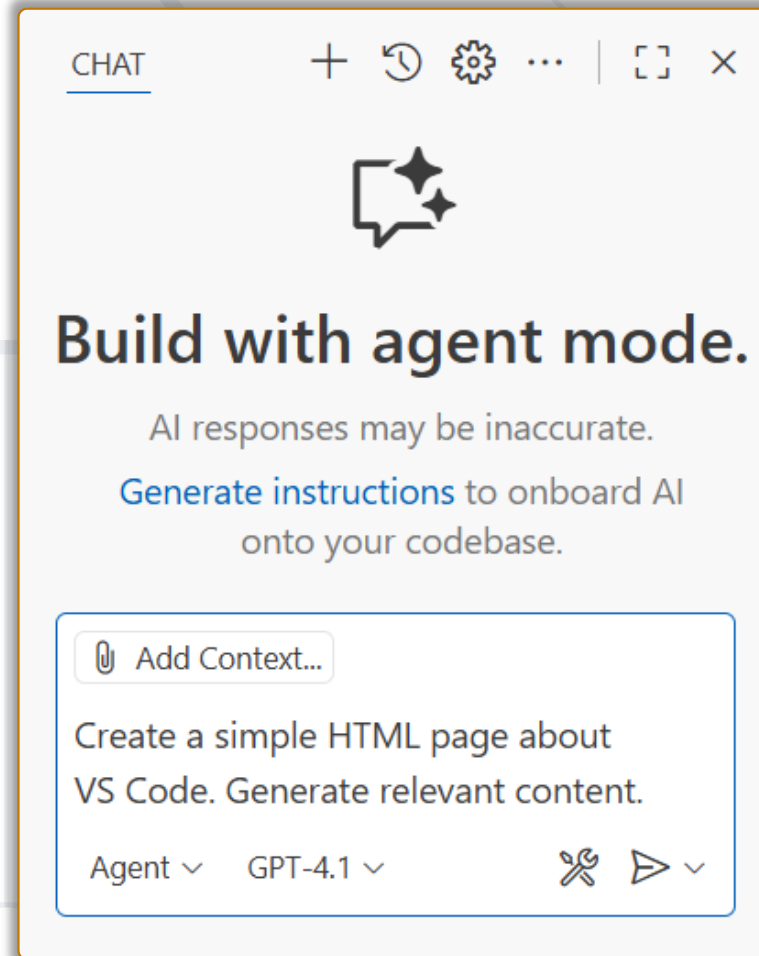


Creating a HTML Page

- Type this sample **prompt** in the GitHub Copilot chat:

Create a **simple HTML** page about VS Code.
Generate relevant content.

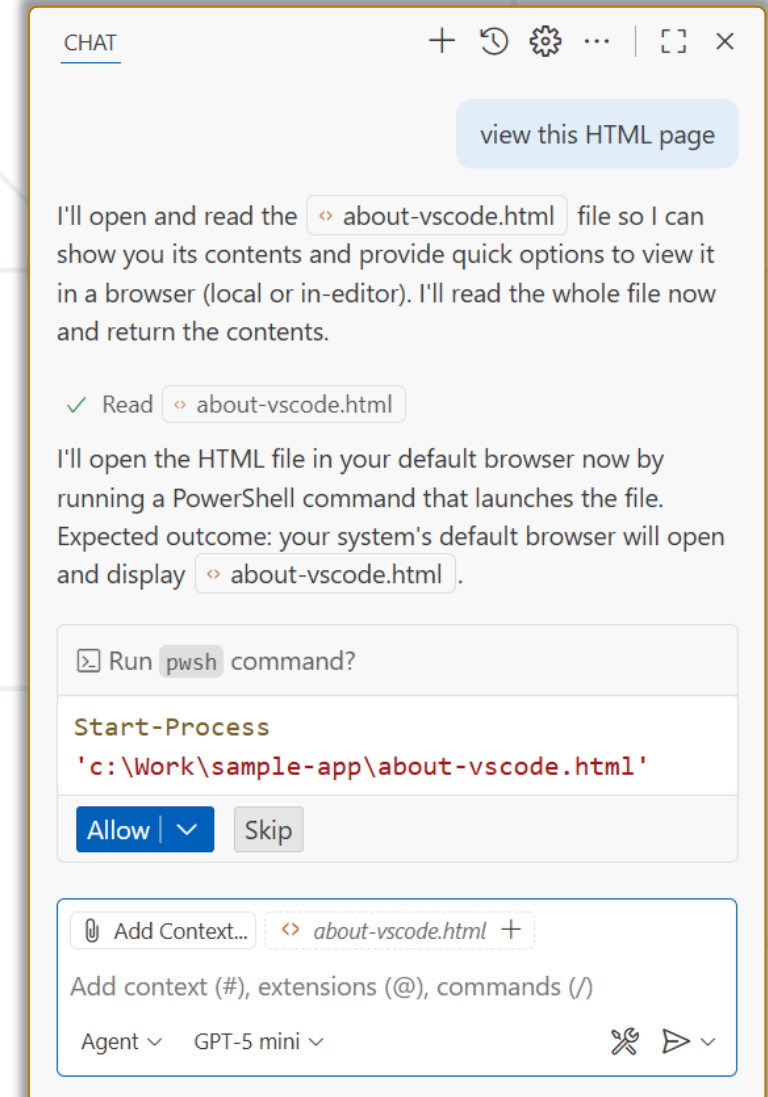
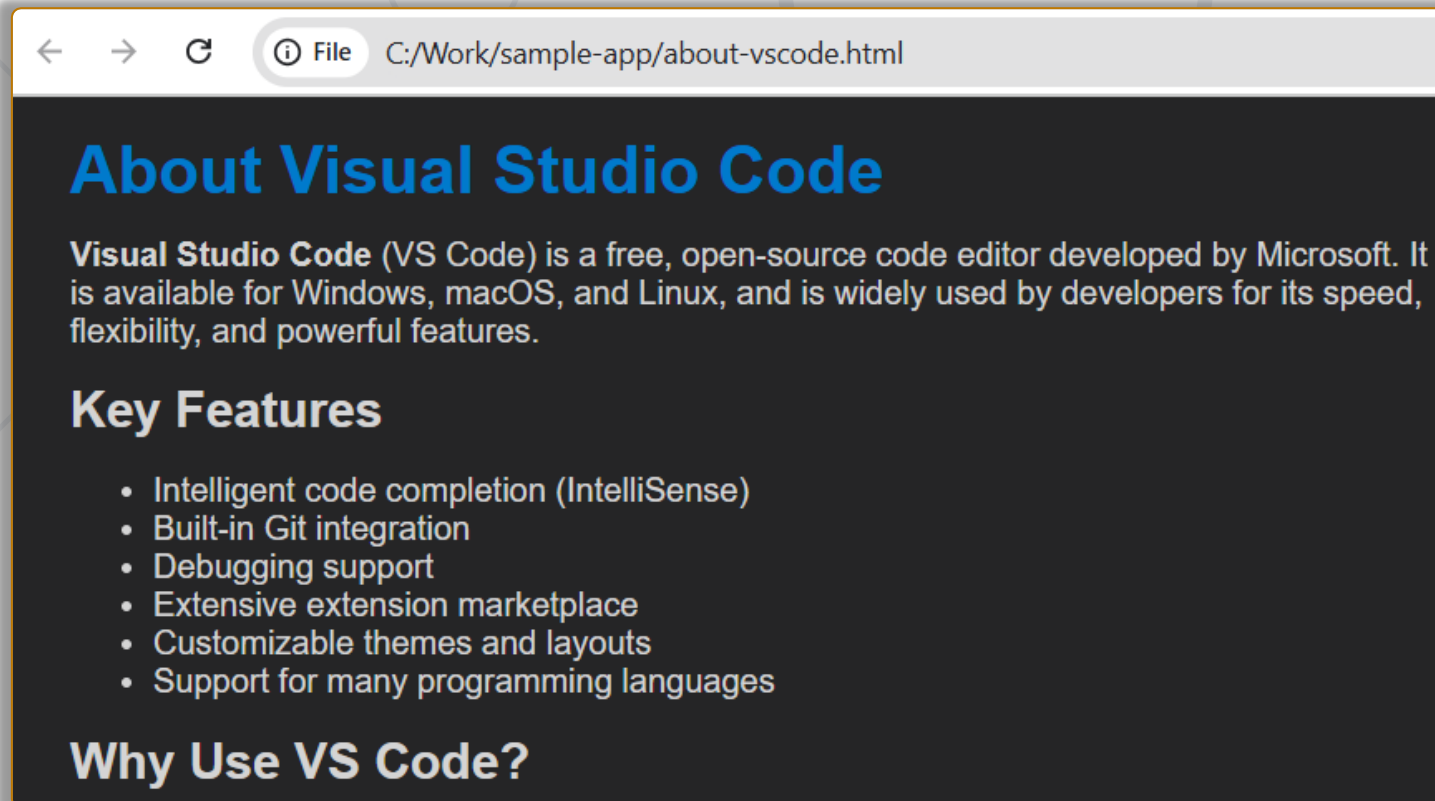
- Wait** for the Web page to be generated
- Accept** the proposed changes

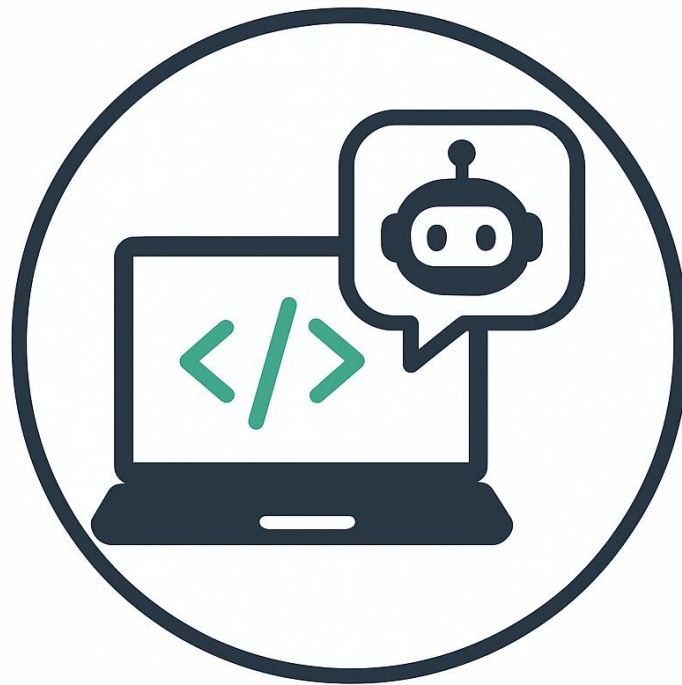


View the Generated HTML Page

- **Open** the page in the Web browser:

View this HTML page



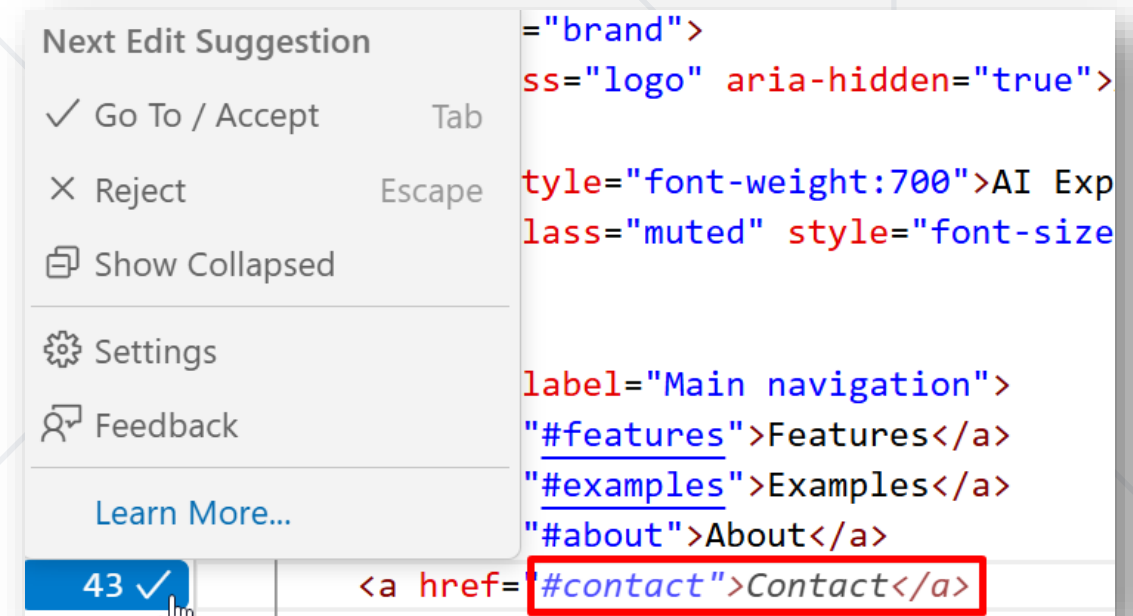


Inline Assistance

Your AI Pair Programmer Inside the Editor

Autocomplete

- **Suggests** code as you type
- **Learns** from file **context** and **patterns**
- **Reduces** boilerplate and repetitive typing
- Works across functions, classes, and full blocks
- **Improves speed** without breaking your flow



- Write a **comment** → Copilot turns it into **code**
- Supports **natural language** instructions
- Great for stubbing new functions quickly
- Can **generate** unit tests based on **requirements**
- **Saves time** on **routine** logic and structure

```
# Print top 5 most common products  
print("\nTop 5 most common products: ")
```

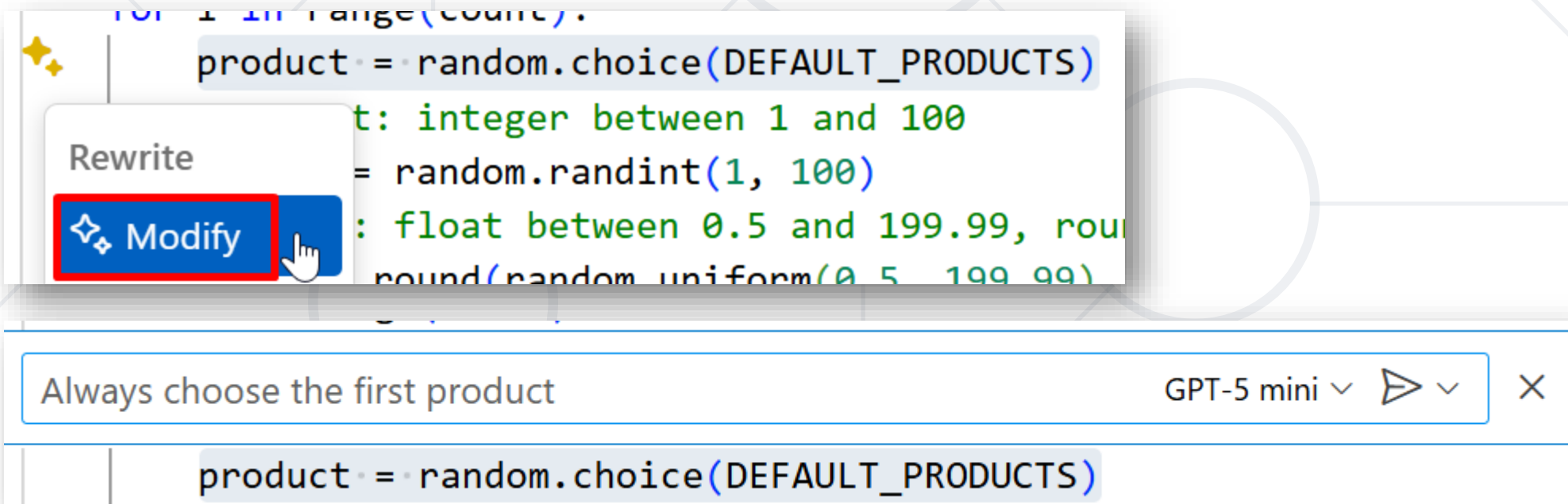
```
# Print top 5 most common products  
print("\nTop 5 most common products: ")  
print(df['product'].value_counts().head(5).to_string())
```

- **Summarizes** selected **code** in plain language
- Helps you **understand unfamiliar logic**
- Useful for **onboarding, debugging, or refactoring**

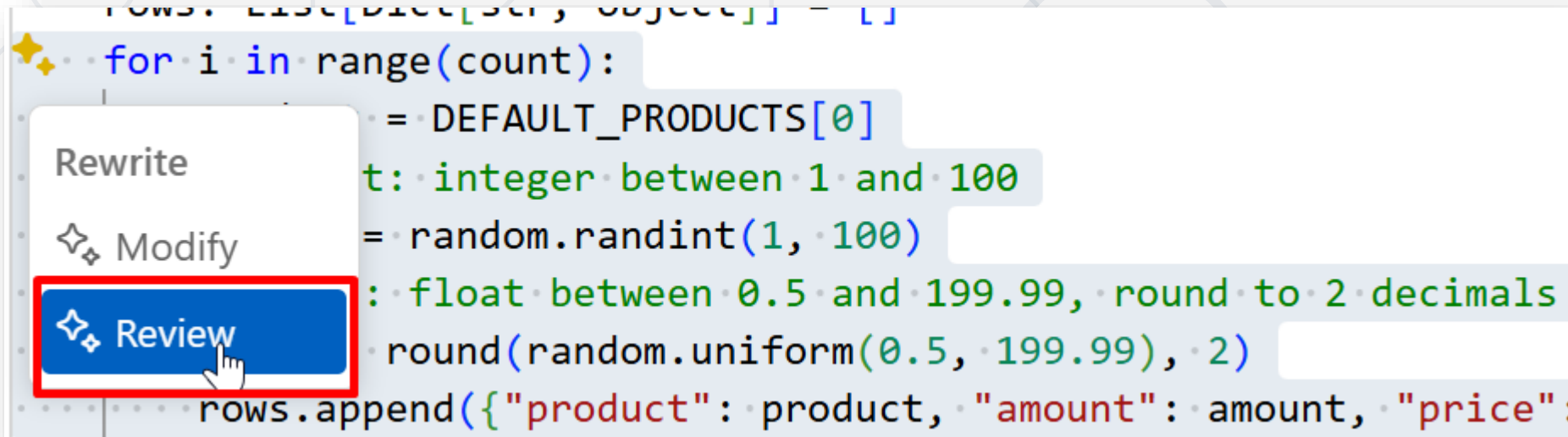
```
const a = document.body.animate(  
  [{ background: from }, { background: to }],  
  { duration: 420, easing: 'ease-in-out', fill: 'for'  
});  
a.onfinish = () => {  
  if (to === lightBg) {  
    root.style.setProperty('--bg', lightBg);  
    root.style.setProperty('--text', '■ #0b1220');  
  } else {  
    root.style.setProperty('--bg', darkBg);  
    root.style.setProperty('--text', '□ #e6eef8');
```

Find All References	Shift+Alt+F12
Find All Implementations	
Show Call Hierarchy	Shift+Alt+H
Add Selection to Chat	
Add File to Chat	
Open Inline Chat	Ctrl+I
Explain	

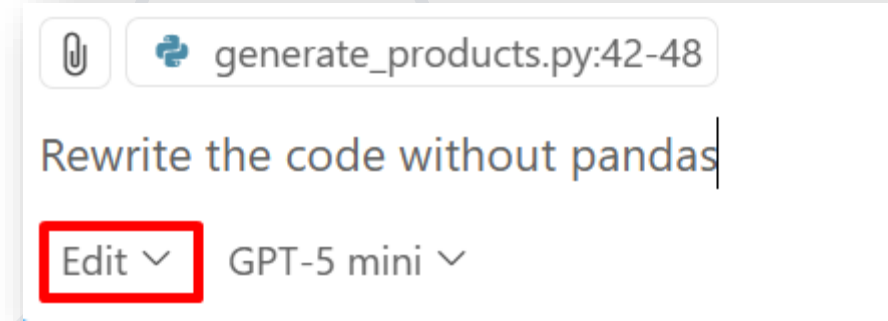
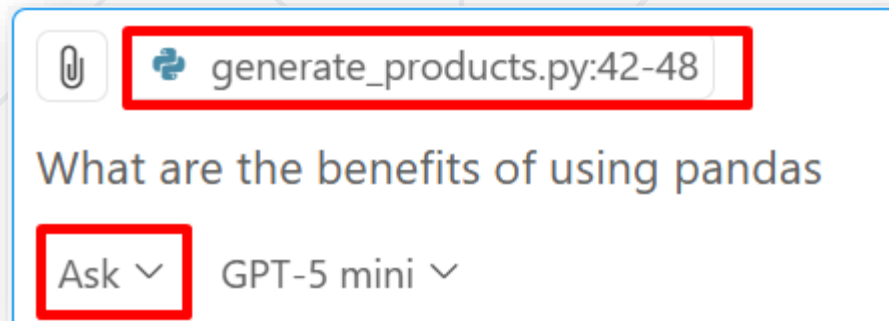
- **Select** code → tell Copilot how to **change** it
- Supports **optimization**, **cleanup**, and **rewrites**
- Can adjust **language**, **style**, or **complexity**



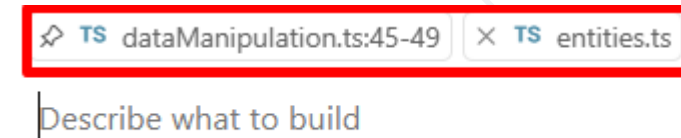
- **Flags** potential **issues** in selected code
- Suggests **improvements** and **best practices**
- Helps **catch bugs** early
- Can focus on **performance**, **readability**, or **security**



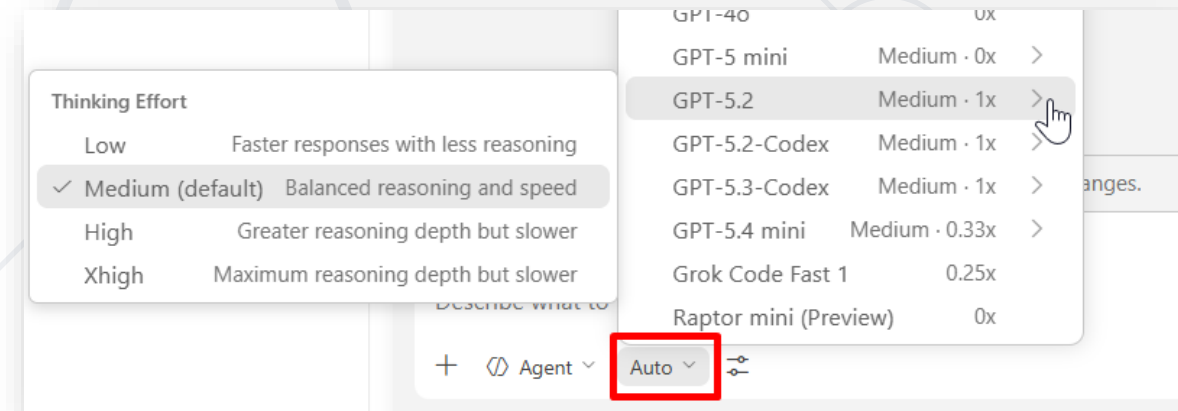
- Ask **questions** about your **code** or **project**
- Get **explanations, examples, or alternatives**
- Works with **selected code** or **entire files**



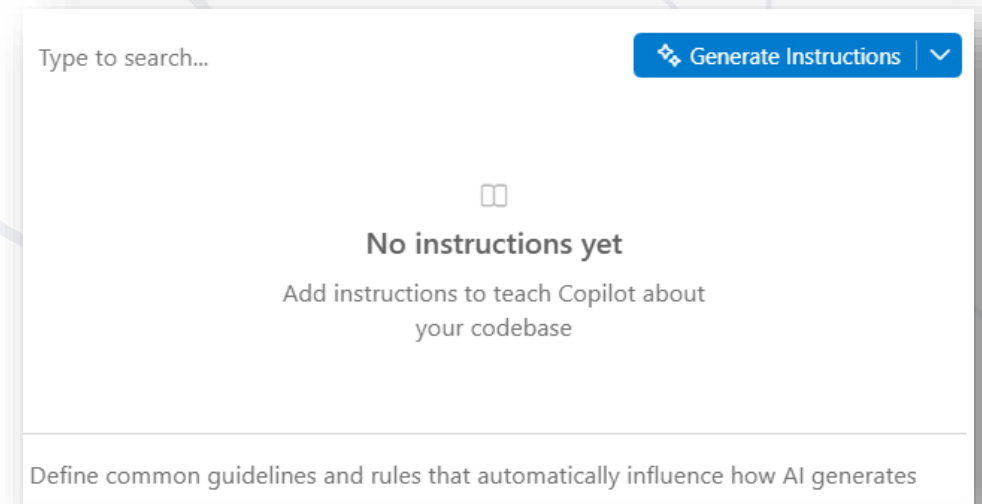
- Better **context** → better AI responses
- Use **#** to add context
- Add **files, folders, symbols**
- VS Code **auto-adds relevant** context
- **Drag & drop** or use **Add Context**



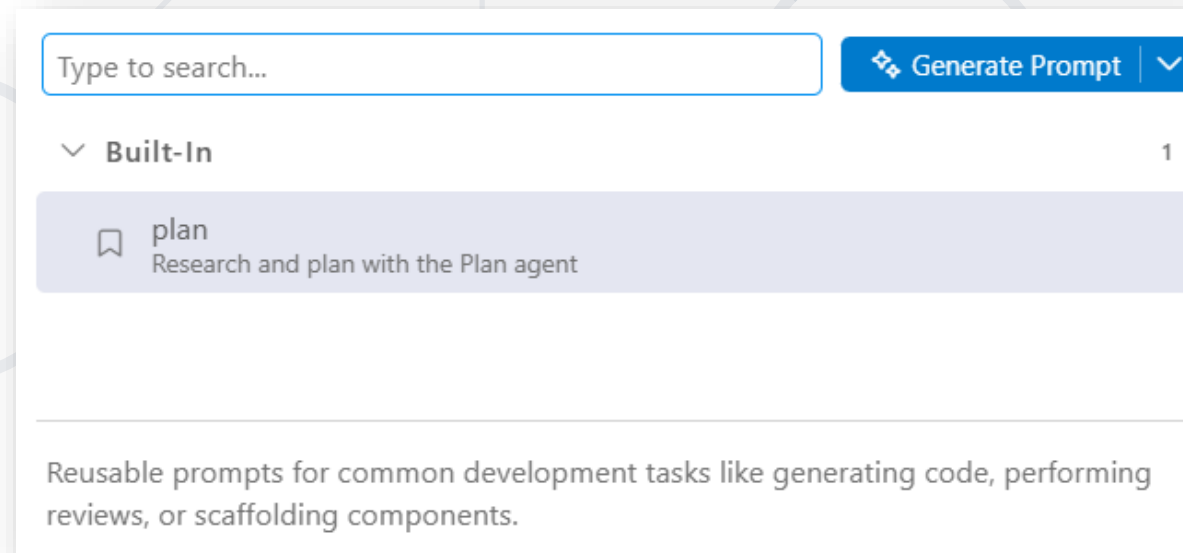
- **Provider Choice:** Toggle between models from OpenAI, Anthropic, and Google Gemini within the chat.
- **Auto-Selection:** Let Copilot dynamically route your request to the best model based on task complexity.
- **Reasoning Models:** Select "Thinking" models for complex architecture and deep debugging.
- **Speed vs. Power:** Choose lightweight models for fast boilerplate and high-performance models for logic-heavy refactoring.



- **Configure** how Copilot behaves
- Set **custom rules** and **preferences**
- Keep style **consistent** across the project
- Acts as a **persistent** system **prompt**
- Helps maintain coding standards



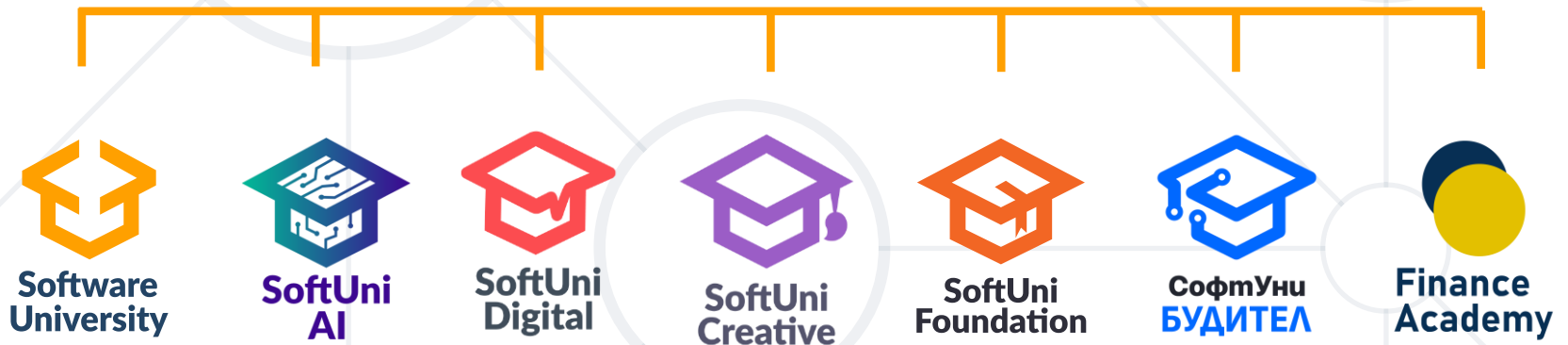
- Save **reusable prompts** directly inside your repo
- Share instructions with the **whole team**
- Useful for **templates** and **recurring tasks**
- Copilot **automatically loads** them in Chat



Questions?



SoftUni





- **Faster** coding with smart **suggestions** and **chat**
- Instantly **generate**, **explain**, and **refactor** code
- **Improve** code **quality** and **consistency** across the project



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